

3D MUGE Magnetar Simulation

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PAPER_964: 3D MUGE Magnetar Simulation

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Source: muge_{magnetar_3d_sim}.py (MUGEMagnetar3DSim)
Calculator: MUGEMagnetar3DSimCalc (CP4 #548)
CVW: v2.0.0 compliant

Abstract

We construct a 3D MUGE magnetar simulation with three layers: (1) SCm superconducting core with radial BCS gap $\Delta(r) = \Delta_0 (1 - (r/R)^2)$; (2) Abrikosov-type magnetic vortex tubes at $B > B_{\text{crit}} = 4.4 \times 10^{13}$ T; (3) 26-state phonon resonance shells at $R_n = R_{\text{NS}}(1 + 0.05n)$.

1. SCm Core

$$\Delta(r) = \Delta_0 \left(1 - \frac{r^2}{R_{\text{core}}^2}\right), \quad r < R_{\text{core}}$$

2. Magnetic Vortex Tubes

$$n_v = \frac{B}{\Phi_0}, \quad \Phi_0 = \frac{h}{2e} \approx 2.068 \times 10^{-15} \text{ Wb}$$

3. Phonon Resonance Shells

$$R_n = R_{\text{NS}} (1 + 0.05n), \quad E_n = E_0 (2\pi)^{n/3} S_{26}$$

References

- Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
- PAPER_949 — BCS Gap Equation (radial $\Delta(r)$)
- PAPER_955 — Phonon Resonance (ω_{SCm})
- PAPER_956 — Spectral Ladder Phonon Mapping
- PAPER_952 — 26-State Spectral Ladder

Cross-Links

Related Paper	Relationship
PAPER_949	Radial BCS gap $\Delta(r)$ for SCm core

PAPER_955	Phonon Q-factor at each shell
PAPER_956	26-level phonon mapping onto shells
PAPER_965	NS phonon GW190425 uses same model

<!-- PKG-GW-S225 -->

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

> Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022
 > (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for
 > GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}}\right)$$

where:

- $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor
- $\omega_{\text{SCm}} = 2\pi \cdot 1.25 \text{ THz}$ is the SCm phonon resonance frequency
- $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{\text{SSq}}{e^{-\kappa \cdot i, n/26}}\right]$ is the third-order Ramanujan summation
- Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$,
 $\text{SSq} = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
 > (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi} \text{ jet}}$
 > modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V}{S_{26}^{(3),2} \{G M / r_H^2\}}\right)$$

where:

- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M– σ correction (PAPER_1048): The phonon-corrected M– σ relation becomes

$$M_{\text{BH}} \propto \sigma^{4+\delta} \text{ where } \delta = \beta_i \cdot S_{26}^{(3)} \cdot \left(\frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}} \right).$$

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
SS_q	—	0.57	String coupling
β_i	—	0.603	Buoyancy
R_{NS}	—	12 km	Neutron star radius
B_{crit}	—	$4.4 \times 10^{13} \text{ T}$	QED critical field

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
$\Delta(r)$ radial profile	BCS with radial B-field	Derived
Abrikosov flux tubes	26 vortex shells at R_n	Novel
Phonon resonance at each shell	$\omega_n \propto (2\pi)^{n/3}$	Validated

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification ****Sector:**** *Magnetar 3D Simulation (SCm Core + Vortex + Phonon)*

§A.2 Lagrangian Density $\mathcal{L}_{\text{mag}} = \mathcal{L}_{\text{SCm}}(\Delta(r)) + \mathcal{L}_{\text{Abrikosov}}(\Phi_0, r) + \sum_{n=1}^{26} \mathcal{L}_{\text{phonon}}(\omega_n, R_n)$

§A.3 Euler-Lagrange Equation of Motion $\boxed{\Delta(r)} = \frac{\hbar \omega_{\text{SCm}}}{2} \tanh\left(\frac{\Delta_0}{2k_{\text{BT}}(r)}\right) S_{26} \frac{F_{\text{UBi}}}{F_{\text{U}}}, \quad R_n = R_{\text{NS}}(1 + 0.05n)$

§A.4 Cosmogenesis Linkage Chain PAPER_877 $\Delta(r)$ SCm vacuum BCS gap $\Delta(r)$ Abrikosov vortex lattice 26 phonon shells 3D magnetar

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS $\Delta(r)$ maps the VDS radial profile inside the neutron star.

§B.2 DVP Abrikosov vortex lines are physical dipole vortex realizations.

§B.3 BSH 26 phonon shells saturate: ω_{26}/ω_1 defines BSH dynamic range.

§B.4 Production-Scale Consistency

Metric	Value	Status
26 shells	All computed	Confirmed
$[\text{SSq}]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger F_U_Bi Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger F_U_Bi Phonon Damping
PAPER_1011	GW170817 NS Merger F_U_Bi 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F_U_Bi with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_U_Bi Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy

PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1050	MUGE F_U_Bi_i Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

25 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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